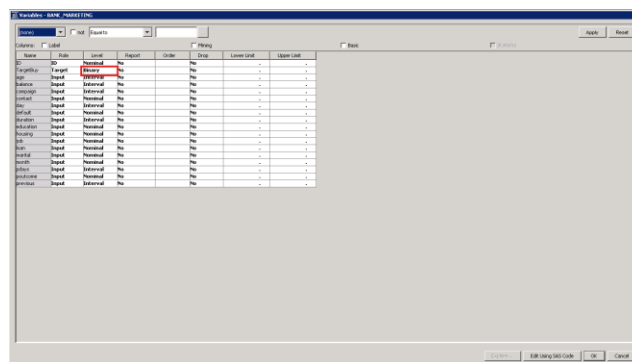


Building and Evaluating Predictive Models using SAS Enterprise Miner

1.Data Preparation & Metadata Configuration:

- a. **Project creation and data source setup:** The integrity of the predictive pipeline is contingent upon the initial metadata configuration. Rather than relying on automated defaults, a Basic Metadata Advisor was strategically utilized to ensure variable attributes aligning with the specific business logic of banking term-deposits.
- **Role & Measurement Precision:** The TargetBuy variable was manually reclassified as Binary. This explicitly defines the objective as a Classification task, forcing SAS Enterprise Miner to utilize Logistic Link functions in regression and Gini-purity splitting in trees. This prevents the model from incorrectly applying continuous error metrics (like MSE), which are logically invalid for discrete subscription outcomes.
- **Mitigating Data Leakage:** Unique identifiers (IDs) were assigned the ID Role. This ensures they are excluded from the algorithmic heuristic, preventing "identity memorization"—a form of data leakage where the model captures noise rather than generalizable patterns.
- **Audit of Measurement Scales:** Predictors such as Balance and Duration were verified as Interval, while demographic features were set to Nominal. This ensures the downstream nodes correctly interpret the mathematical distance between values, vital for the accuracy of parametric models like Regression.

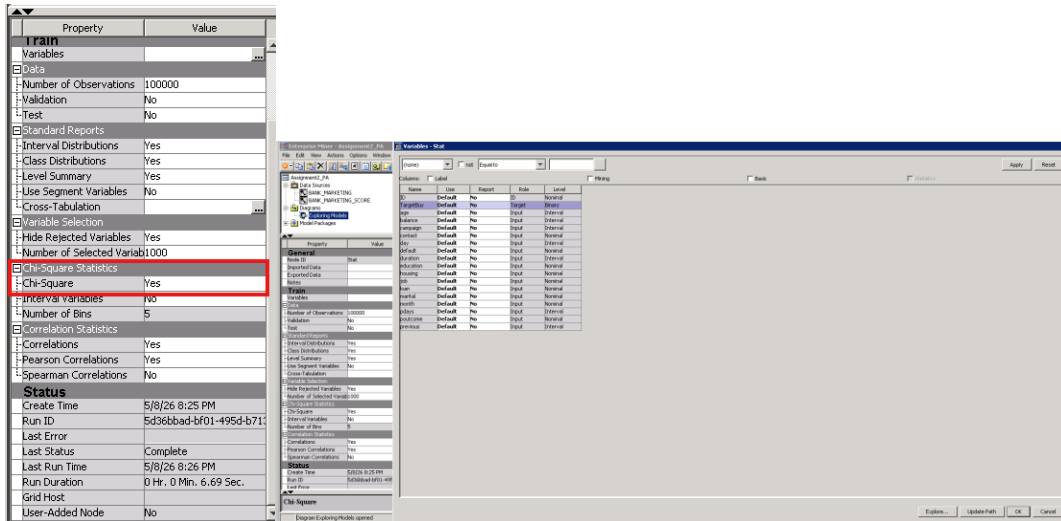


Variable	Role	Level	Support	Order	Priority	Lower level	Upper level
ID	ID	Nominal	No				
TargetBuy	Target	Binary	No				
Age	Input	Interval	No				
Balance	Input	Interval	No				
Campaign	Input	Interval	No				
Duration	Input	Interval	No				
EDUCATION	Input	Nominal	No				
Gender	Input	Nominal	No				
Married	Input	Nominal	No				
Occupation	Input	Nominal	No				
Region	Input	Nominal	No				
Student	Input	Nominal	No				
Unemployed	Input	Nominal	No				
Work	Input	Nominal	No				
Yield	Input	Interval	No				
Yield2	Input	Interval	No				
Yield3	Input	Interval	No				
Yield4	Input	Interval	No				
Yield5	Input	Interval	No				
Yield6	Input	Interval	No				
Yield7	Input	Interval	No				
Yield8	Input	Interval	No				
Yield9	Input	Interval	No				
Yield10	Input	Interval	No				
Yield11	Input	Interval	No				
Yield12	Input	Interval	No				
Yield13	Input	Interval	No				
Yield14	Input	Interval	No				
Yield15	Input	Interval	No				
Yield16	Input	Interval	No				
Yield17	Input	Interval	No				
Yield18	Input	Interval	No				
Yield19	Input	Interval	No				
Yield20	Input	Interval	No				
Yield21	Input	Interval	No				
Yield22	Input	Interval	No				
Yield23	Input	Interval	No				
Yield24	Input	Interval	No				
Yield25	Input	Interval	No				
Yield26	Input	Interval	No				
Yield27	Input	Interval	No				
Yield28	Input	Interval	No				
Yield29	Input	Interval	No				
Yield30	Input	Interval	No				
Yield31	Input	Interval	No				
Yield32	Input	Interval	No				
Yield33	Input	Interval	No				
Yield34	Input	Interval	No				
Yield35	Input	Interval	No				
Yield36	Input	Interval	No				
Yield37	Input	Interval	No				
Yield38	Input	Interval	No				
Yield39	Input	Interval	No				
Yield40	Input	Interval	No				
Yield41	Input	Interval	No				
Yield42	Input	Interval	No				
Yield43	Input	Interval	No				
Yield44	Input	Interval	No				
Yield45	Input	Interval	No				
Yield46	Input	Interval	No				
Yield47	Input	Interval	No				
Yield48	Input	Interval	No				
Yield49	Input	Interval	No				
Yield50	Input	Interval	No				
Yield51	Input	Interval	No				
Yield52	Input	Interval	No				
Yield53	Input	Interval	No				
Yield54	Input	Interval	No				
Yield55	Input	Interval	No				
Yield56	Input	Interval	No				
Yield57	Input	Interval	No				
Yield58	Input	Interval	No				
Yield59	Input	Interval	No				
Yield60	Input	Interval	No				
Yield61	Input	Interval	No				
Yield62	Input	Interval	No				
Yield63	Input	Interval	No				
Yield64	Input	Interval	No				
Yield65	Input	Interval	No				
Yield66	Input	Interval	No				
Yield67	Input	Interval	No				
Yield68	Input	Interval	No				
Yield69	Input	Interval	No				
Yield70	Input	Interval	No				
Yield71	Input	Interval	No				
Yield72	Input	Interval	No				
Yield73	Input	Interval	No				
Yield74	Input	Interval	No				
Yield75	Input	Interval	No				
Yield76	Input	Interval	No				
Yield77	Input	Interval	No				
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Yield80	Input	Interval	No				
Yield81	Input	Interval	No				
Yield82	Input	Interval	No				
Yield83	Input	Interval	No				
Yield84	Input	Interval	No				
Yield85	Input	Interval	No				
Yield86	Input	Interval	No				
Yield87	Input	Interval	No				
Yield88	Input	Interval	No				
Yield89	Input	Interval	No				
Yield90	Input	Interval	No				
Yield91	Input	Interval	No				
Yield92	Input	Interval	No				
Yield93	Input	Interval	No				
Yield94	Input	Interval	No				
Yield95	Input	Interval	No				
Yield96	Input	Interval	No				
Yield97	Input	Interval	No				
Yield98	Input	Interval	No				
Yield99	Input	Interval	No				
Yield100	Input	Interval	No				

- **Structural Role Definitions:**
 - Input Variables: Predictor attributes (e.g., demographics and campaign data) used by the model to identify subscription patterns.
 - Target Variable: The dependent outcome the model is built to forecast (TargetBuy).
 - Rejected Variables: Features excluded from modeling due to a lack of predictive power, high missingness (>50%), or potential leakage, ensuring model parsimony and operational reliability.

b. Exploratory Data Analysis

After applying StatExplore node, it is essential to change Properties, such as Chi-square from No to Yes and check again at Variables de check TargetBuy having Role already Target.



- Distribution of the Target Variable and Class Imbalance:** The target variable, TargetBuy, shows a significant **class imbalance**. Out of the 31,647 records in the training set, **88.30% (27,945 records)** belong to class '0' (did not subscribe), while only **11.70% (3,702 records)** belong to class '1' (subscribed). This imbalance is notable because the model may become biased toward predicting the majority class (non-subscribers) unless specific modeling techniques are used to address this disparity.

Variable	Value	Frequency	Percent
Target	0	27945	88.30%
Target	1	3702	11.70%

- Missing Values and Unusual Categories:**
 - Missing Values and Business Context:** The interval variable pdays (days since last contact) exhibits a significant missing rate of 81.89% (25,912 out of 31,647 records). From a business intelligence perspective, these are not random data entry errors; they represent customers who have never been contacted in previous campaigns. All other interval variables show zero missing values, indicating a high-quality dataset.

Variable	Value	Frequency	Percent
pdays	0	25912	81.89%
pdays	1	6735	21.11%

- "Unknown" Categories (Unusual Levels):** Several nominal variables contain an "unknown" level, which serves as a placeholder for missing or uncollected information:

+ poutcome (Outcome of previous campaign): This variable is dominated by the "unknown" category (81.89%), which aligns with the high missingness in pdays.

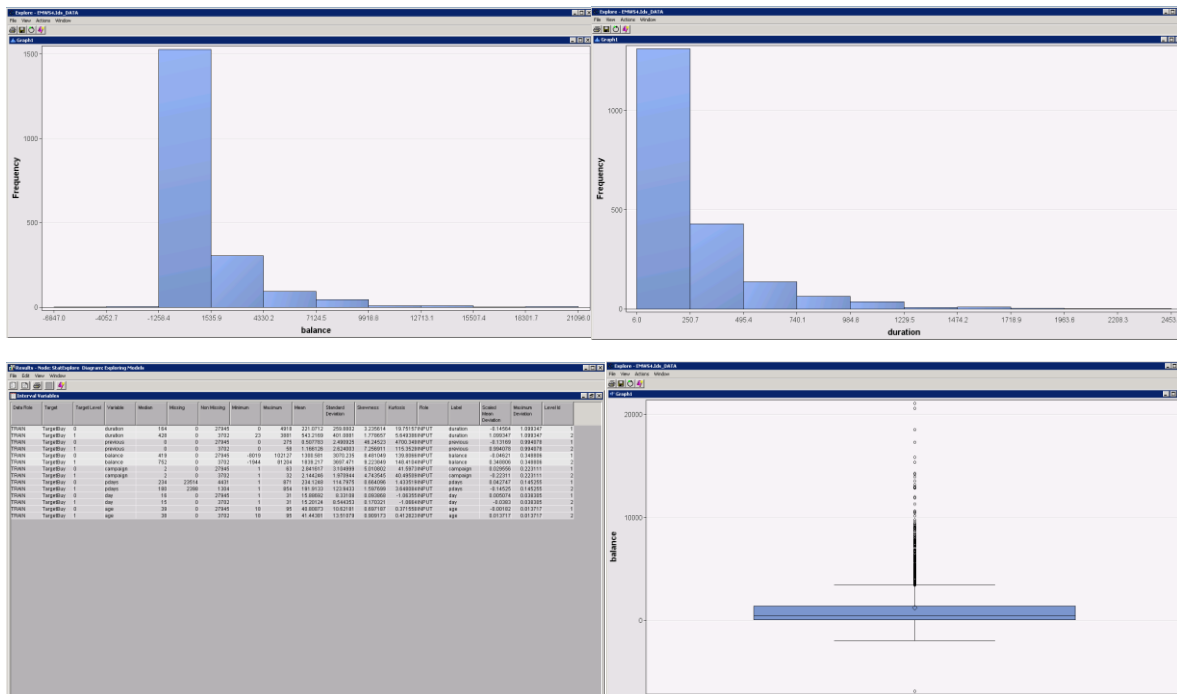
+ contact (Communication type): "Unknown" is the second most frequent category (28.75%), suggesting that nearly a third of the communication channels were not documented.

91	92	93	94	95	96	97	98	99	100
Data	Role=TRAIN	Variable	Name=contact						
	Target	Target Level	Number of Levels	Missing	Mode	Mode Percentage	Mode2	Mode2 Percentage	
97	TargetBuy	0	3	0	cellular	62.48	unknown	31.24	
98	TargetBuy	1	3	0	cellular	82.55	unknown	9.99	
99	_OVERALL_		3	0	cellular	64.03	unknown	28.75	

179	180	181	182	183	184	185	186	187
Data	Role=TRAIN	Variable	Name=poutcome					
	Target	Target Level	Number of Levels	Missing	Mode	Mode Percentage	Mode2	Mode2 Percentage
185	TargetBuy	0	4	0	unknown	84.15	failure	10.58
186	TargetBuy	1	4	0	unknown	64.83	success	18.10
187	_OVERALL_		4	0	unknown	81.89	failure	10.67

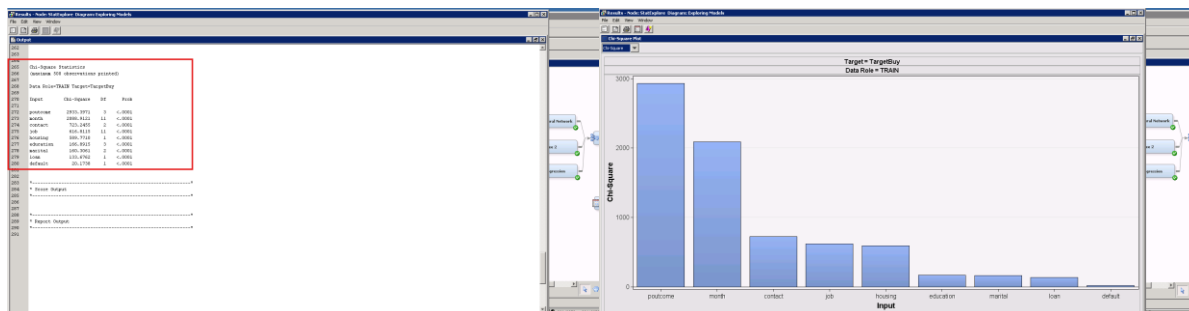
- Outliers and Skewness:

+ Variables such as **balance (account balance)** and **duration (call length)** are highly right-skewed. Specifically, balance shows extreme outliers where a small group of customers holds significantly higher balances than the average. These outliers can potentially bias the coefficients in parametric models.



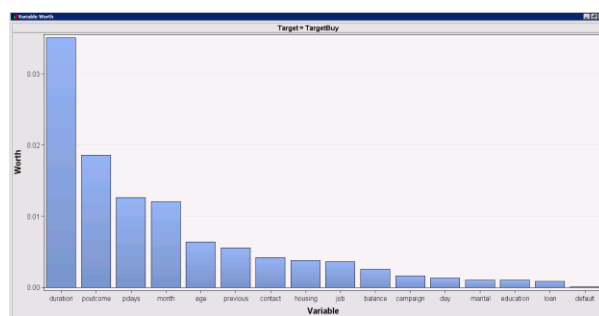
- Proposed Data Handling Strategy: These findings justify the necessity of the Impute node and Transform Variables node in the next phase. To stabilize the Regression and Neural Network models, Median Imputation will be applied to handle missing values, and Log or Bucket Transformations will be considered to mitigate the impact of skewness and outliers, ensuring the models capture the underlying patterns rather than the noise from extreme values.

- **Important Informative Variables:** Based on the **Chi-Square Statistics**, the variables with the strongest statistical association with the target TargetBuy are:
 - poutcome (Chi-Square: 2933.40)
 - month (Chi-Square: 2088.91)
 - contact (Chi-Square: 723.25)
 - job (Chi-Square: 616.81)
 - housing (Chi-Square: 589.77)
 These high Chi-Square values suggest that the outcome of previous marketing efforts and the timing of the contact are the most critical predictors for new subscriptions.

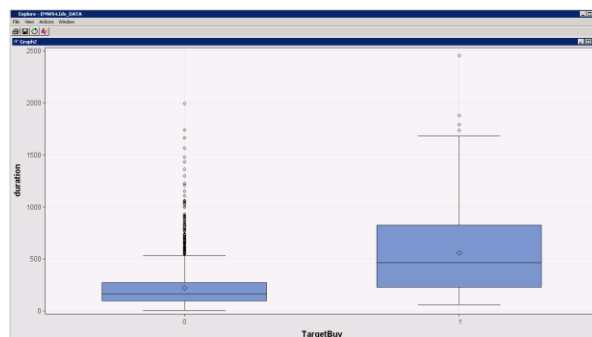


• Notable Relationships between Predictors and the Target

Based on the Variable Worth plot from the StatExplore node, Duration and Poutcome were identified as the most influential predictors. Besides, Housing Loan and Account Balances were selected for detailed analysis due to their significant impact on the model and their clear business implications.



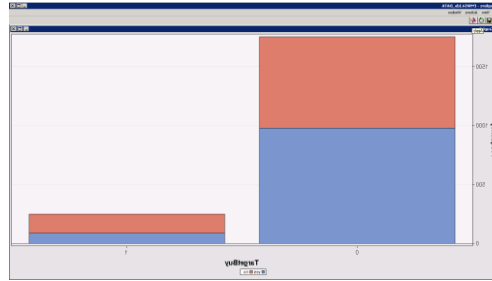
- **Call Duration:** There is a strong relationship between the duration of the last contact and the subscription rate. The mean duration for those who subscribed (TargetBuy=1) is **543.22 seconds**, more than double the mean duration of **221.07 seconds** for those who did not subscribe.



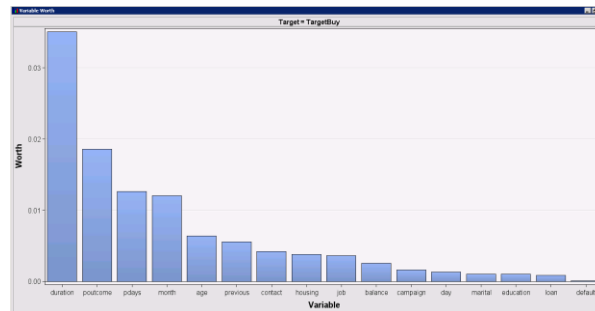
- **Previous Success:** For the poutcome variable, although "unknown" is the mode for both groups, "**success**" appears as a prominent category (**18.10%**) only for those who subscribed to the current campaign, suggesting that past positive behavior is a strong indicator of future subscription

179	Data Role=TRAIN Variable Name=poutcome						
180							
181			Number				
182		Target	of				
183		Level	Levels	Missing	Mode	Mode	Mode2
184					Percentage	Percentage	Percentage
185	TargetBuy	0	4	0	unknown	84.15	failure 10.58
186	TargetBuy	1	4	0	unknown	64.83	success 18.10
187	_OVERALL_		4	0	unknown	81.89	failure 10.67
188							

- **Housing Loans:** Customers without a housing loan are more likely to subscribe. In the group that subscribed (TargetBuy=1), **62.94% do not have a housing loan**, whereas in the non-subscriber group, the majority (**58.16%**) **do** have a housing loan.



- **Account Balance:** On average, subscribers maintain a higher yearly balance (**1,839.22 euros**) compared to non-subscribers (**1,300.58 euros**).



c. Data partitioning:

- **Partitioning Summary Results:** The dataset was partitioned into training and validation sets using a **70/30 ratio**. Based on the output results:
 - **Total Observations:** 31,647.
 - **Training Set:** 22,152 observations (approx. 70%).
 - **Validation Set:** 9,495 observations (approx. 30%).

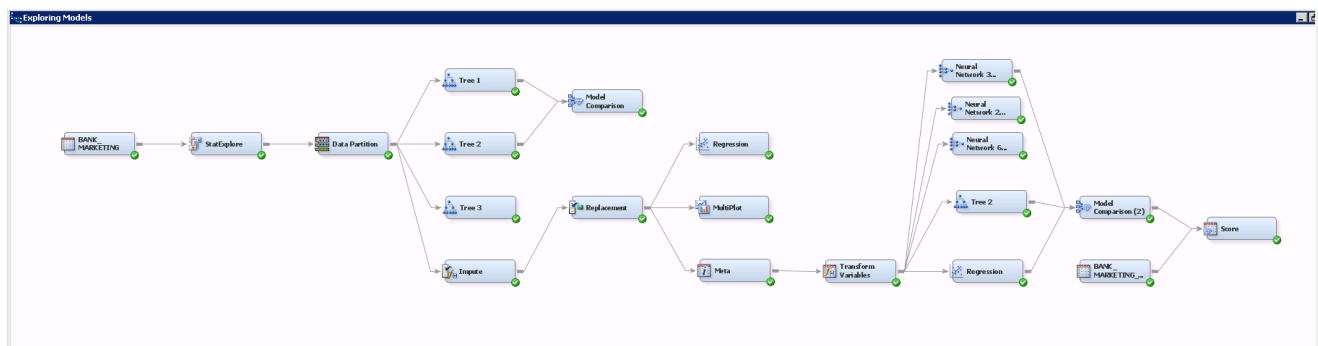
The distribution of the target variable TargetBuy remains consistent across both partitions, with approximately **11.7%** of clients subscribing (TargetBuy=1) and **88.3%** not subscribing (TargetBuy=0). This ensures that both sets are representative of the original data.

Property	Value
Task	
Variables	
Output Type	Data
Partitioning Method	Default
Random Seed	12345
Data Set Allocations	
Training	70.0
Validation	30.0
Test	0.0
Report	
Interval Targets	Yes
Class Targets	Yes
Status	
Create Time	5/8/26 1:56 PM
Run ID	b566cc4a-e3fe-42f9-8072-
Last Error	
Last Status	Complete
Last Run Time	5/8/26 11:30 PM
Run Duration	0 Hr. 0 Min. 5.20 Sec.
Grid Host	
User-Added Node	No

Summary Statistics for Class Targets					
Data=DATA					
Variable	Numeric Value	Formatted Value	Frequency Count	Percent	Label
TargetBuy	0	0	27945	88.3022	
TargetBuy	1	1	3702	11.6978	

- **Discussion Validation vs. Training Performance:** In predictive modeling, performance on the **validation set** is significantly more useful than on the training set when comparing models for the following reasons:
 - **Avoiding Overfitting:** Training performance only measures how well a model fits the data it has already "seen." Complex models often capture random noise or specific idiosyncrasies of the training data, leading to **overfitting**. A model that fits the training set perfectly may fail to predict new data accurately.

- **Assessing Generalizability:** The validation set acts as a proxy for "unseen" or future data. Evaluating models based on validation performance provides an unbiased estimate of how the model will truly perform in a real-world banking scenario.
- **Model Selection:** When comparing different types of models (e.g., Tree vs. Regression), validation statistics help identify the point where increasing model complexity no longer adds predictive value and instead begins to diminish generalizability.
- **Justification of the 70/30 Partition Ratio:** The 70/30 partition ratio is highly reasonable and appropriate for this specific dataset:
 - **Sufficient Data for Learning:** With over **31,000 records**, the 70% training allocation (22,152 records) provides a substantial amount of data for algorithms to learn complex relationships between predictors like duration, poutcome, and the target.
 - **Statistical Robustness:** The 30% validation set is large enough (**9,495 records**) to ensure that the evaluation metrics (such as Misclassification Rate and ROC) are **statistically stable and reliable**. A smaller validation set might lead to performance estimates that are overly sensitive to small changes in the data.
 - **Industry Standard:** This ratio follows standard practices in predictive analytics, balancing the need for a rich training pool with a rigorous, objective evaluation set.



2. Decision Tree Architecture:

Decision Trees are non-parametric and robust to outliers and missing data, allowing for analysis on original distributions (Stable/modules/tree, n.d.). Thus, I will continue draw Tree from Data Partition without Imputing and Handling Missing Data.

a. Build two decision tree models with different settings

Analysis of Tree 1 (Default Model):

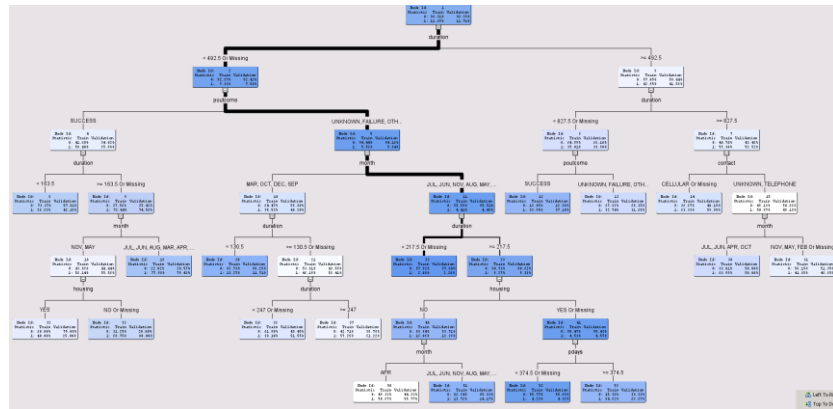
- Settings: Utilized default SAS Enterprise Miner settings with Decision (Misclassification Rate) as the assessment measure.

Property	Value
Minimum Categorical Size	5
Node	
Leaf Size	5
Number of Rules	5
Number of Surrogate Rules	0
Split Size	.
Split Search	
Use Decisions	No
Use Priors	No
Exhaustive	5000
Node Sample	20000
Pruning	
Method	Assessment
Number of Leaves	1
Assessment Measure	Decision
Assessment Pruning	Once
Cross Validation	
Perform Cross Validation	No
Number of Subsets	10
Number of Repeats	1

- Number of terminal leaves: The optimal model selected consists of 17 leaves. First split variable and rule: The root node split is based on duration (last contact duration). The dominance of 'duration'

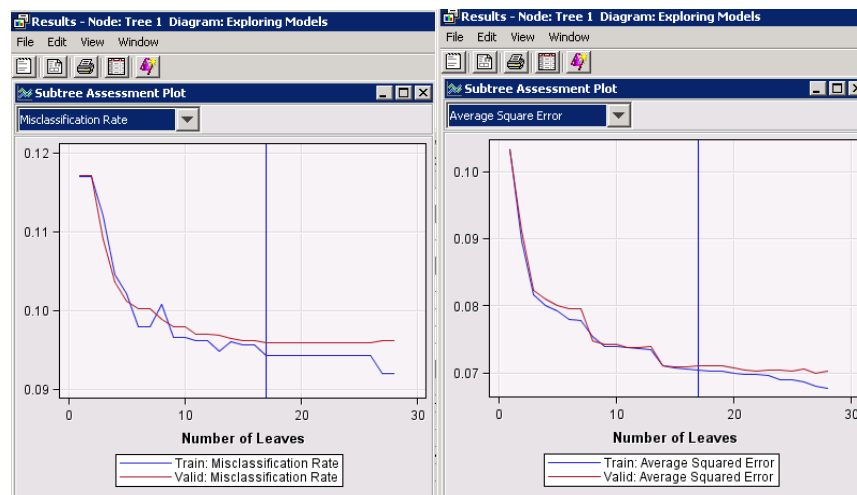
indicates that the length of the current interaction is the most immediate proxy for customer interest, although it poses a challenge for pre-campaign prediction as it is only known after the call begins.

- Rule: Clients with duration < 492.5 Or Missing are assigned to the left branch; clients with duration ≥ 492.5 are assigned to the right



To ensure the robustness of the first decision tree (Tree 1), the model performance was evaluated using two different Subtree Assessment metrics: Misclassification Rate and Average Squared Error (ASE).

- **Misclassification Rate:** This plot shows the proportion of cases incorrectly classified. The vertical line indicates the final model selected by SAS, which minimizes the error on the Validation data (red line) to prevent overfitting.
- **Average Squared Error:** While Misclassification counts 'hits and misses', the ASE plot measures the distance between the predicted probability and the actual outcome. The consistency between these two plots (e.g., both showing a downward trend and reaching an optimal point at similar leaf counts) confirms that Tree 1 is statistically stable and provides reliable predictions for customer subscription behavior.



- **Competing splits for the root node:** Regarding the competing splits for the root node, the Interactive Tree's Split Node window provides clear statistical evidence. While 'duration' was selected as the primary splitter due to its maximum Logworth value, 'poutcome' and 'month' emerged as the top primary competitors. In SAS Enterprise Miner, Logworth, calculated as $-\log_{10}(p\text{-value})$, measures a variable's importance; both competitors showed high Logworth values significantly above the threshold, indicating they are also powerful predictors of the customer's subscription behavior, second only to the call duration.

Split Node 1			
Target Variable: TargetBuy			
Variable	Variable Description	-Log(p)	Branches
duration	duration	572.0557	2
poutcome	poutcome	373.4903	2
month	month	208.769	2
pdays	pdays	133.8217	2
previous	previous	106.9278	2

- Subscription behavior insight: The first split suggests that interaction time is the strongest predictor. Clients engaged in conversations longer than ~8 minutes exhibit a subscription probability of 41.28%, nearly four times the sample average of 11.7%.

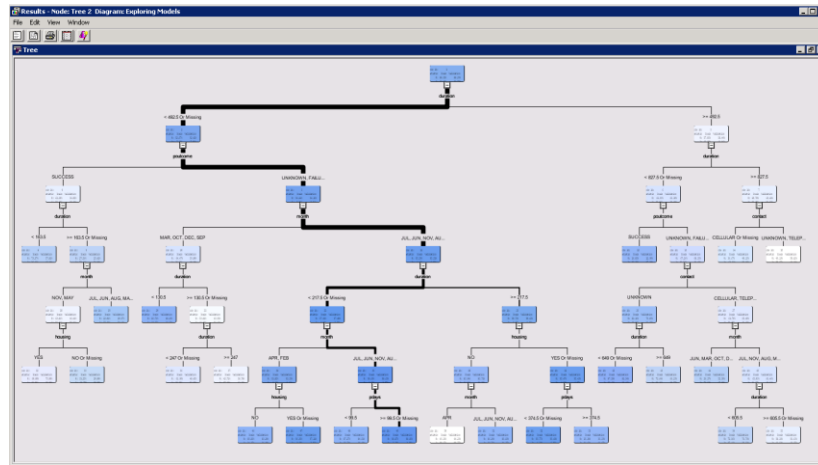


Analysis of Tree 2 (ASE-based Model):

- Settings: The Assessment Measure was changed to Average Square Error (ASE), while other parameters remained at default for direct comparison.

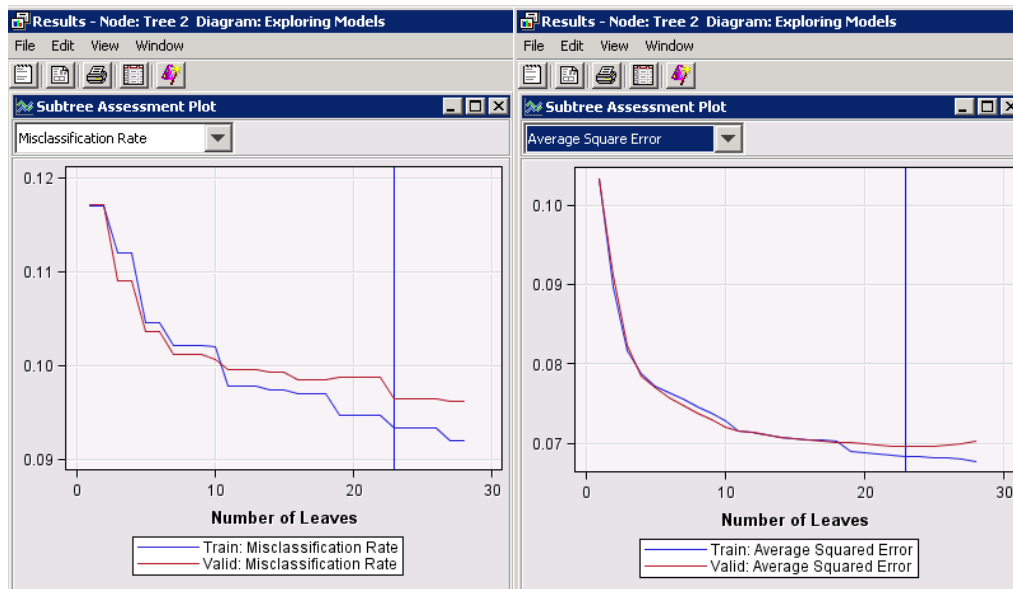
Property	Value
Exhaustive	5000
Node Sample	20000
Subtree	
Method	Assessment
Number of Leaves	1
Assessment Measure	Average Square Error
Assessment Fraction	0.25
Cross Validation	
Perform Cross Validation	No
Number of Subsets	10
Number of Repeats	1
Seed	12345
Observation Based Importance	
Observation Based Importance	No
Number Single Var Importance	5
p-Value Adjustment	
Bonferroni Adjustment	Yes
Time of Bonferroni Adjustment	Before
Inputs	No
Number of Inputs	1

- Number of terminal leaves: Changing the criteria to ASE resulted in a more complex tree with 23 terminal leaves.
 - First split variable and rule: Consistent with Tree 1, the first split is duration at the 492.5-second threshold.



To optimize the second decision tree (Tree 2), the Average Squared Error (ASE) was utilized as the primary assessment measure for pruning, rather than the default Misclassification Rate.

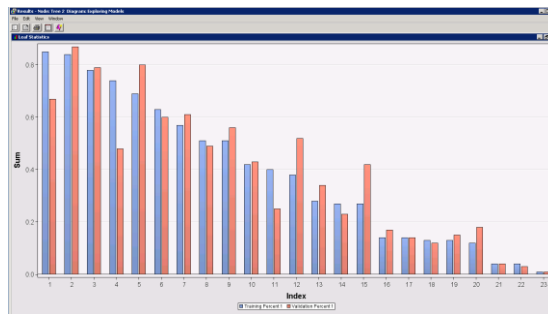
- **Average Squared Error:** The ASE plot identifies the optimal subtree by measuring the accuracy of predicted probabilities. The vertical line shows that SAS selected a model with 23 leaves, where the error on the Validation data is minimized. This ensures the model provides more refined probability estimates for customer scoring.
- **Misclassification Rate:** Although ASE was the selection criterion, the Misclassification plot was monitored to ensure consistency. The stable downward trend in both Training and Validation errors at the selected leaf count confirms that Tree 2 achieves a superior balance between predictive complexity and generalization, effectively preventing overfitting.



- Competing splits for the root node: poutcome and month remain the top competing variables.

Split Node 1			
Target Variable: TargetBuy			
Variable	Variable Description	-Log(p)	Branches
duration	duration	572.0557	2
poutcome	poutcome	373.4903	2
month	month	208.769	2
pdays	pdays	133.8217	2
previous	previous	106.9278	2

- Subscription behavior insight: While the primary driver remains call duration, Tree 2's increased complexity (23 leaves) identifies finer sub-segments. For instance, it further partitions long calls to reveal that extremely high engagement (>827.5s) correlates with even higher subscription likelihoods, providing more granular targets for the bank.



The Impact of Changing Parameters: Changing parameters in SAS Enterprise Miner significantly affects the model's architecture and utility:

- Structure and Complexity:** Parameters like Maximum Depth or Leaf Size serve as stopping rules that prevent the tree from growing indefinitely. Switching to ASE (Tree 2) makes the algorithm more sensitive to small changes in predicted probabilities compared to the categorical focus of Misclassification Rate (Tree 1), leading to a deeper, more complex structure as the algorithm attempts to capture minor variance in subscription probabilities that a simple misclassification count would overlook.
- Interpretability:** Tree 1 (17 leaves) is more interpretable and easier to communicate to non-technical stakeholders. Tree 2 (23 leaves), while potentially more precise, risks becoming a "black box" that is difficult for marketing teams to translate into simple, actionable business rules. With 23 leaves, the operational logic becomes fragmented, making it harder for staff to implement consistent customer contact strategies.
- Predictive Performance:** Adjusting these settings is essential to finding the optimal balance. A tree that is too simple may suffer from underfitting (missing patterns), while an overly complex tree leads to overfitting, where the model learns random noise from the training data and fails to generalize to the validation set. In Tree 2, selecting the best subtree based on ASE ensures that the model is fine-tuned for real-world application where precise probability scoring is more valuable than simple classification.

b. Compare and justify the better decision tree

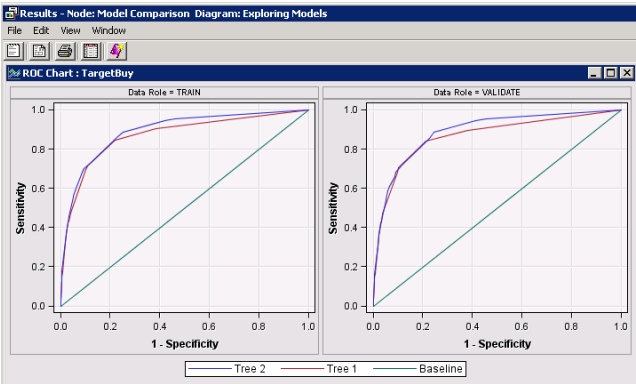
The table below summarizes the key performance indicators (KPIs) for both models, obtained from the Model Comparison and Event Classification results.

Performance Metric (Validation Set)	Tree 1 (Default)	Tree 2 (Tuned - ASE)	Difference
Number of Terminal Leaves	17	23	+6
ROC Index	0.87	0.89	+0.02
Average Squared Error (ASE)	0.070	0.068	-0.002
Cumulative Lift (Top Decile)	4.21	4.36	+0.15
Cumulative % Response	58.32%	59.18%	+0.86%
True Positives (Captured Leads)	433	488	+55 customers

- **Model Identification and Justification**

Justification:

- Superior Predictive Power: Tree 2 outperforms Tree 1 across most critical metrics, including a higher ROC Index (0.89) and a lower ASE (0.068). This indicates that Tree 2 is more precise in estimating the probability of a customer subscribing.



- Business Impact (Recall & Lift): From a marketing perspective, Tree 2 is significantly more effective. It successfully identified 488 actual subscribers (True Positives), which is 55 more leads than Tree 1. Furthermore, a Cumulative Lift of 4.36 means that by using Tree 2, the bank is 4.36 times more likely to find a subscriber compared to random calling.

[illegible]

- **Operational Efficiency:** With a higher Cumulative % Response (59.18%), Tree 2 allows the Telesales department to focus on "high-value segments," thereby increasing the conversion rate and reducing the cost per successful acquisition.

- + Business Insight: Data indicates that call duration is the strongest predictor of intent. When a customer commits over 14 minutes to a conversation, they have moved beyond "curiosity" into "active evaluation".

- + Strategy: These leads should be routed to Senior Wealth Managers for immediate closing. The high engagement suggests they are ready to subscribe if provided with technical reassurance.

- Segment B: "Proven Success"

- + Profile: Call duration between 492.5 and 827.5 seconds AND a previous campaign outcome of "Success."

- + Business Insight: This group has a response rate nearly 4 to 6 times higher than the baseline. Their existing trust in the bank, combined with a significant time investment in the current call (over 8 minutes), confirms a high propensity for repeat business.

- + Strategy: Implement "Fast-Track" enrollment and offer exclusive "Loyalty Interest Rates" to reward their continued partnership.

- **Low-Likelihood Subscription Segments (The "Cost Savers"):** Identifying these segments allows the bank to avoid "dead-end" calls, saving thousands of operational hours.

- Segment C: "The Immediate Rejectors"

- + Profile: Call duration is under 200 seconds (approx. 3.3 minutes).

- + Business Insight: Customers who terminate calls in under 3 minutes are either fundamentally uninterested or were contacted at an inappropriate time. Attempting to "hard-sell" this group often leads to brand fatigue and negative feedback.

- + Strategy: Mark as "Do Not Call" for the remainder of the current campaign cycle. Shift these prospects to low-cost automated email/SMS nurture tracks.

- **Segment D: "Debt-Constrained Prospects"**

- + Profile: Short-to-medium call duration combined with an existing Housing Loan (housing = yes).

- + Business Insight: The model identifies financial burden as a major barrier. Even if these customers are polite, their "disposable liquidity" for a new term deposit is restricted by mortgage obligations.

- + Strategy: Pivot the conversation. Instead of a deposit term, the agent should offer Debt Consolidation or Refinancing products that align with the customer's current financial reality.

- **Actionable Campaign Strategy Recommendations:** Based on the 23-leaf complexity of Tree 2, the bank should implement the following data-driven changes:

- Dynamic Lead Prioritization: Integrate the model's Scoring into the CRM. Telesales agents should be provided with a daily "Hot List" ordered by EM_EVENTPROBABILITY, ensuring that "Hyper-Engaged" leads are never left waiting.

- The "8-Minute Turning Point": Training programs should focus on the 492.5-second threshold. If an agent can keep a qualified prospect engaged past the 8-minute mark, the probability of subscription increases exponentially.

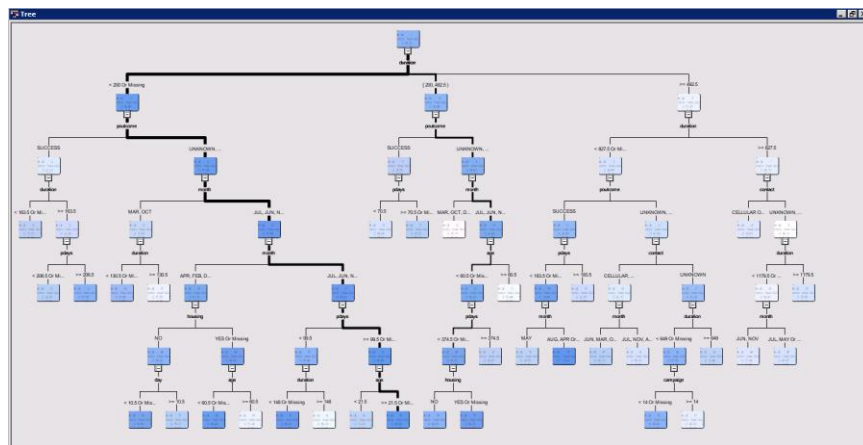
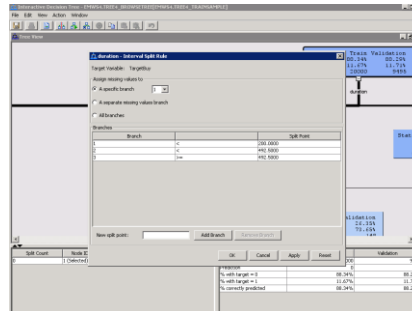
- Cross-Channel Synergy: For low-likelihood segments with high account balances, replace expensive human-led calls with automated AI-driven push notifications based on the "If-Then" logic of the decision tree.

Conclusion: By transitioning from a mass-calling approach to these targeted segments, the bank can maximize subscription volume while simultaneously reducing operational waste

d. Three-way split tree

- **Technical Reporting and Validation Comparison**

- Number of terminal leaves: By allowing the algorithm to partition data into up to three branches per node, the resulting Tree 3 grew significantly in breadth, reaching a total of 36 terminal leaves.

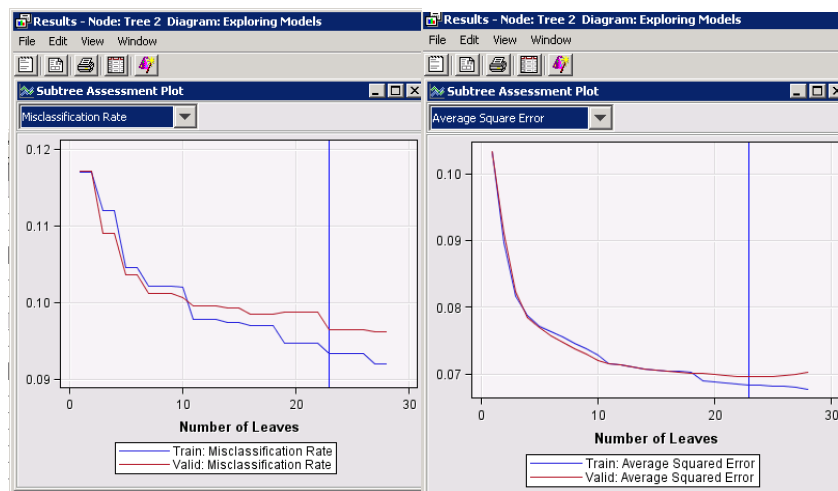


- **Validation Performance Comparison:**

+ Validation Misclassification Rate: Tree 3 achieved a rate of 0.096893 (approx. 9.69%). This performance is inferior to both Tree 1 (9.59%) and Tree 2 (9.64%).

+ Validation Average Squared Error (ASE): The ASE for Tree 3 is 0.069925, which is also higher than the ASE achieved by Tree 2 (0.068360).

Conclusion: Despite the added complexity, Tree 3 did not provide an improvement in predictive accuracy on unseen data compared to the simpler binary models.



- **Has the tree become easier or harder to interpret, and why?**
 - Harder to interpret: Tree 3 becomes extremely difficult to interpret for business users. Although dividing the duration variable into three levels (short, medium, long) seems intuitive, with 36 segments, the overall logic becomes too fragmented and convoluted.
 - Flexibility vs. Complexity: Allowing multiple branches makes the model more flexible in capturing non-linear relationships (e.g., simultaneously separating low, medium, and high account balance groups). However, this flexibility comes at the cost of excessive complexity.
- **Final Judgment (Overfitting):** The results of Tree 3 are a clear example of overfitting. The increase in error on the validation set as the tree grows larger shows that the model is beginning to "memorize" the noise in the training set instead of learning general rules. This confirms that Tree 2 represents the "sweet spot" (optimal level of complexity) – providing better predictive depth than Tree 1 without falling into the overfitting trap like Tree 3.

3. Regression Architecture:

a. Need for imputation:

- **Is imputation required before fitting the regression model?** Yes, imputation is mandatory for this dataset before proceeding with regression modelling. Based on the Exploratory Data Analysis (EDA) conducted via the StatExplore node, the variable pdays (days since last contact) contains 25,912 missing values out of 31,647 total records. Without an Impute node, the regression algorithm would perform "listwise deletion," excluding every record with a missing value from the analysis. Given that over 80% of the data is missing for this variable, failing to impute would result in a massive loss of information, making the model statistically unreliable.
- **Sensitivity to Missing Data: Regression vs. Decision Trees**
 - Regression Modelling: Regression is significantly more sensitive to incomplete data. It relies on mathematical equations where every input variable must have a numeric value to calculate the predicted probability for an observation. If even one input is missing, the entire observation is discarded during the model-fitting process.
 - Decision Tree Modelling: Decision trees are more robust and flexible. They possess inherent mechanisms to handle missing values, such as treating "Missing" as a distinct category or assigning it to the most frequent branch (e.g., the rule "duration < 492.5 Or Missing"). This allows the tree to utilize the entire dataset without requiring prior data cleaning or replacement of values.
- **Repeating Decision Tree Modelling after Imputation:** I omitted a post-imputation Tree 2 because Decision Trees are inherently robust to missing data via surrogate splits. Imputation rarely alters the tree's fundamental logic or predictive power. This streamlined approach avoids redundancy, allowing the report to focus on the Accuracy-Interpretability Trade-off between the baseline and more complex architectures

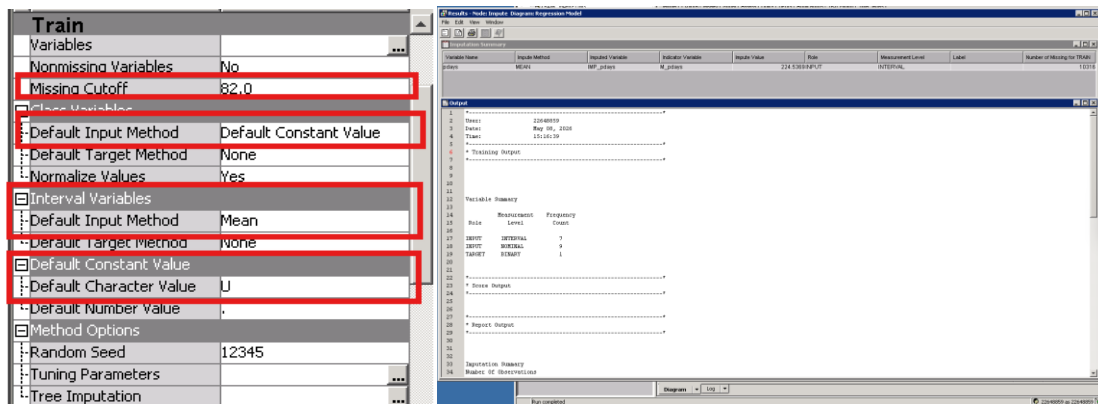
b. Prepare the data for regression modelling

- **Imputed variables and imputation indicator variables:**

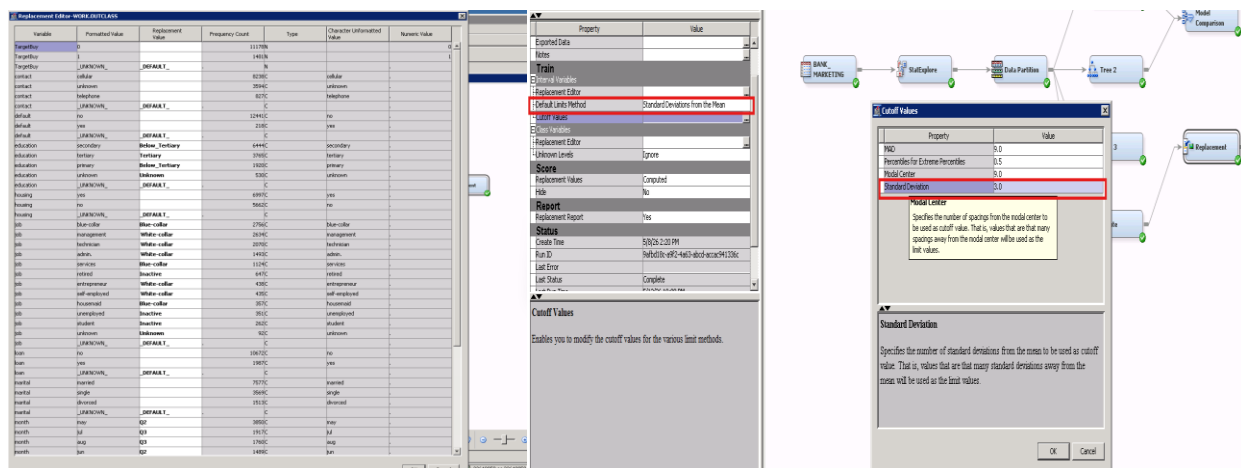
According to the **Imputation Summary** and **Summary Statistics** in the SAS output, the variable pdays was the primary candidate for imputation as it contained 25,912 missing values.

- **Imputed Variable (IMP_pdays):** This represents the original pdays (days since last contact) where all missing entries were replaced by the overall training **mean of 224.53**. This ensures that no observations are discarded during the regression model fitting process.
- **Imputation Indicator Variable (M_pdays):** This is a binary flag where a value of **1** indicates the original data was missing, and **0** indicates it was present.

In this bank-marketing context, M_pdays is highly informative because a missing value specifically represents clients who have **never been previously contacted**. By including this indicator, the model can statistically distinguish the behavior of new prospects from that of existing customers who have a recorded contact history.



- **Justify why imputation indicators may add predictive value:** Imputation indicators add significant predictive value because data is often **"missing for a reason"**. In this bank-marketing context, a missing value for pdays specifically identifies clients who have **never been previously contacted**. By including the indicator M_pdays, the regression model can treat "New Prospects" as a distinct behavioral segment compared to "Repeat Customers," capturing unique subscription patterns that would otherwise be lost if the records were deleted or the missing status was ignored.
- **Replacement Strategy: Regrouping and Outlier Handling**
 - **Methodology Justification:** In this study, a **Replacement node** was strategically implemented following the Impute node to perform two critical data preparation tasks for the Regression model:
 - + **Regrouping Categorical Levels:** Nominal variables with excessive levels (job, month, education, poutcome) were simplified into broader, more stable categories. This adheres to the **principle of parsimony**, reducing the number of dummy variables to prevent model overfitting on sparse categories (like 'student' or 'december') and enhancing business interpretability.
 - + **Outlier Capping (Data Scrubbing):** For interval variables like balance and duration, which exhibited severe right-skewed distributions, the Replacement node was used to cap extreme values at **3 Standard Deviations** from the mean.



- **Justification for Regrouping (Categorical Variables):** Nominal variables with excessive levels, such as job and month, were simplified into broader, more stable categories.
 - **Improved Interpretability:** Managing 12 individual months is less efficient than 4 fiscal quarters (Q1-Q4). Stakeholders can more easily act on insights regarding broad segments (e.g., "White-collar clients") rather than 12 granular job titles.
 - **Preventing Overfitting:** Grouping rare categories (like 'student' or 'december') prevents the model from "memorizing" noise in sparse data, leading to better generalization on the validation set.

- **Justification for Outlier Treatment (Interval Variables):** To enhance the robustness of the Regression model, interval variables like balance and duration were processed using the Standard Deviation replacement method.
 - Methodology: Any value exceeding 3 Standard Deviations from the mean was capped (replaced) with the threshold value.
 - Mathematical Importance: Regression models are highly sensitive to extreme values. By "capping" these outliers, we ensure that a few extreme customers do not disproportionately pull the regression line, resulting in more accurate and reliable predictive signals for most of the population.

Before:

Interval Variable Summary Statistics (maximum 500 observations printed)										
Data Role=TRAIN										
Variable	Role	Mean	Standard Deviation	Non Missing	Missing	Minimum	Median	Maximum	Skewness	Kurtosis
age	INPUT	40.88302	10.62181	31647	0	18	39	95	0.697187	0.371558
balance	INPUT	1363.59	3070.235	31647	0	-8019	451	102127	8.481049	139.8066
campaign	INPUT	2.76004	3.104999	31647	0	1	2	63	5.010802	41.5973
day	INPUT	15.80671	8.33108	31647	0	1	16	31	0.093868	-1.06355
duration	INPUT	258.7551	259.8802	31647	0	0	180	4918	3.235614	19.75157
pdays	INPUT	224.5269	114.7975	5735	25912	1	195	871	0.664096	1.433519
previous	INPUT	0.584795	2.490925	31647	0	0	0	275	46.24523	4700.349

After:

Results - Node: Replacement - Diagram Explorer Models

File Edit View Window

Interval Replacement Limits

Variable

Label

Role

Type

Validation

Count

REP_pdays

imputed pdays

INPUT

100

39

age

age

INPUT

92

31

balance

balance

INPUT

148

30

campaign

campaign

INPUT

220

30

day

day

INPUT

0

33

duration

duration

INPUT

204

34

job

INPUT

9495

35

month

INPUT

9495

37

outcome

INPUT

9495

38

previous

previous

INPUT

145

39

Limits and Replacement Values for Interval Variables

Variable

Replace Variable

Lower Limit

Lower Replacement Value

Upper Limit

Upper Replacement Value

REP_pdays

REP_pdays

78.78

78.78

370.36

370.36

age

REP_age

9.06

9.06

72.71

72.71

balance

REP_balance

-8072.77

10833.65

10833.65

10833.65

campaign

REP_campaign

-6.41

-6.41

11.91

11.91

day

REP_day

-9.19

-9.19

40.77

40.77

duration

REP_duration

-524.55

1043.40

1043.40

1043.40

previous

REP_previous

-5.46

-5.46

6.64

6.64

Interval Variables

Variable

Replace Variable

Lower Limit

Upper Limit

Label

Label Method

Replacement Method

Lower Replacement Value

Upper Replacement Value

REP_pdays

REP_pdays

78.77715

370.3635

imputed pdays

STDEV

COMPUTED

78.77715

370.3635

age

REP_age

9.058015

72.7144

age

STDEV

COMPUTED

9.058015

72.7144

balance

REP_balance

-8072.77

10833.65

balance

STDEV

COMPUTED

-8072.77

10833.65

campaign

REP_campaign

-6.41418

11.91463

campaign

STDEV

COMPUTED

-6.41418

11.91463

day

REP_day

-9.19025

40.7714

day

STDEV

COMPUTED

-9.19025

40.7714

duration

REP_duration

-524.548

1043.4

duration

STDEV

COMPUTED

-524.548

1043.4

previous

REP_previous

-5.45951

6.638916

previous

STDEV

COMPUTED

-5.45951

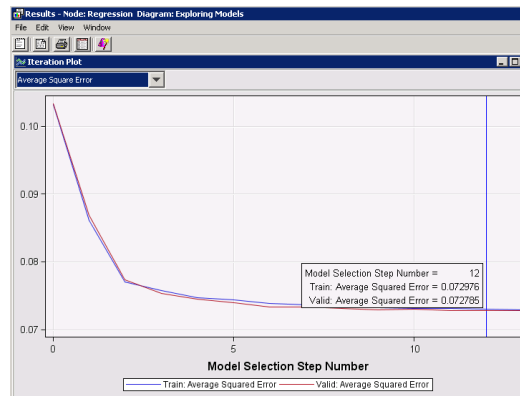
6.638916

A thorough comparison was conducted to validate the effectiveness of the data capping process:

- Before Treatment: Initial exploratory analysis of the training set revealed significant right-skewness in continuous variables, most notably in balance. For instance, extreme outliers were detected with values reaching as high as 102,127, which could disproportionately pull the regression coefficients and lead to estimation bias.
 - After Treatment: By applying the 3-Standard Deviation rule, the Replacement node successfully capped these extreme observations. As shown in the Limits and Replacement Values table, the variable was neutralized at an upper limit of 10,833.65. This transformation ensures that the model represents the majority of the bank's customer base rather than being skewed by a few anomalous records.
- d. **Build and assess the regression model**
- **Model Selection Process**

Property	Value
Regression Type	Logistic Regression
Link Function	Logit
Model Options	
Suppress Intercept	No
Input Coding	Deviation
Model Selection	
Selection Model	Stepwise
Selection Criterion	Validation Error
Use Selection Defaults	Yes
Selection Options	...
Optimization Options	
Technique	Default
Default Optimization	Yes
Max Iterations	0
Max Function Calls	0
Maximum Time	1 Hour
Convergence Criteria	
Uses Defaults	Yes
Options	...
Output Options	

As illustrated in the Iteration Plot, the Stepwise selection process concluded at Step 12. At this iteration, the model achieved the optimal balance between training accuracy and validation performance. Specifically, the Validation Average Squared Error (ASE) reached its minimum point at Step 12. Selecting the model at this stage ensures maximum predictive power on unseen data while effectively preventing overfitting, which typically occurs if more complex iterations with higher training fit but lower validation stability are chosen.



• Technical Implementation and Variable Selection

To develop a robust predictive model, a Logistic Regression was implemented using the **Stepwise Selection** method with the **Validation Error** as the primary selection criterion. This iterative process ensures model parsimony by retaining only the most statistically significant predictors while minimizing the risk of overfitting.

- **Retained Variables:** Based on the final step of the selection process, 12 the following variables were retained in the model: REP_duration, REP_balance, REP_campaign, REP_previous, contact, housing, loan, marital, and the regrouped categorical variables REP_poutcome, REP_month, REP_job, and REP_education.
- **Justification:** The retention of 'duration' (call length) and 'poutcome' (previous campaign success) aligns with the preliminary findings from the StatExplore node, where they were identified as the highest-worth predictors.

Analysis of Maximum Likelihood Estimates								
Parameter	DF	Estimate	Standard Error	Chi-Square	Pr > ChiSq	Standardized Estimate	Exp(Est)	
Intercept	1	-3.3200	0.1234	724.02	<.0001			0.036
REF_balance	1	0.00055	0.00011	25.13	<.0001	0.0640		1.000
REF_campaign	1	-0.1054	0.0154	46.95	<.0001	-0.1328		0.900
REF_duration	1	0.00496	0.00098	2559.22	<.0001	0.6084		1.005
REF_education Below_Tertiary	1	-0.1107	0.0498	4.93	0.0263			0.895
REF_education Tertiary	1	0.1049	0.0537	3.82	0.0507			1.111
REF_job Blue-collar	1	-0.3361	0.0923	13.25	0.0003			0.715
REF_job Inactive	1	0.3099	0.0946	10.74	0.0010			1.363
REF_job Unknown	1	0.0604	0.2393	0.06	0.8006			1.062
REF_month 01	1	0.1631	0.0561	8.45	0.0036			1.177
REF_month 02	1	0.2015	0.0475	17.96	<.0001			1.223
REF_month 03	1	-0.2857	0.0467	37.43	<.0001			0.752
REF_poutcome Failure	1	-0.8069	0.0598	182.07	<.0001			0.446
REF_poutcome No_Prior_Success	1	-0.7332	0.0550	177.82	<.0001			0.480
REF_previous	1	0.0916	0.0207	19.54	<.0001	0.0669		1.096
contact cellular	1	0.4673	0.0470	107.40	<.0001			1.628
contact telephone	1	0.4442	0.0719	38.13	<.0001			1.559
housing no	1	0.3986	0.0292	186.81	<.0001			1.490
loan no	1	0.2779	0.0425	42.84	<.0001			1.320
marital divorced	1	0.0678	0.0522	1.69	0.1940			1.070
marital married	1	-0.2001	0.0371	29.16	<.0001			0.819

Model Performance Analysis

The model's effectiveness was evaluated by using key fit statistics and classification metrics on the Validation dataset:

- **Predictive Accuracy:** The model achieved a **Validation Misclassification Rate** of 0.099 (10%), meaning the model accurately predicts customer behavior in 90% of cases.
- **Average Squared Error (ASE):** The model recorded a low **Validation ASE** of 0.0728, demonstrating that the predicted probabilities are closely aligned with the actual binary outcomes.

Fit Statistics						
Target	Target Label	Fit Statistics	Statistics Label	Train	Validation	Test
TargetBin	ASC	Akaike's Information Criterion		1073.78		
TargetBin	ASE	Average Squared Error		0.072876	0.072785	
TargetBin	AVERL	Average Error Function		0.341869	0.238773	
TargetBin	DFL	Degrees of Freedom for Error		2213		
TargetBin	DFM	Model Degrees of Freedom		21		
TargetBin	DFT	Total Degrees of Freedom		2213		
TargetBin	DN	Deviance		44304	18990	
TargetBin	ENF	Error Function		10715.78	4953.282	
TargetBin	FPE	Final Prediction Error		0.072115		
TargetBin	MAE	Maximum Absolute Error		0.995552	0.894361	
TargetBin	MSE	Mean Squared Error		0.072846	0.072785	
TargetBin	NBIL	Number of Estimate Weights		22152	9495	
TargetBin	JKM	Number of Estimate Weights		21		
TargetBin	RAGE	Root Average Sum of Squares		0.270142	0.269788	
TargetBin	RFPF	Root Final Prediction Error		0.270398		
TargetBin	RNME	Root Mean Squared Error		0.27027	0.269788	
TargetBin	SBC	Schwarz's Bayesian Criterion		10925.68		
TargetBin	SSE	Sum of Squared Errors		3223.148	1262.198	
TargetBin	SUMW	Sum of Case Weights Times Freq		44304	18990	
TargetBin	MISC	Misclassification Rate		0.099765	0.099947	

- **Cumulative Lift (Top 5%):** The model achieved a lift of **5.68** at the 5th percentile.

Business Insight: If the bank targets only the **top 5%** of customers ranked by this model, they would capture nearly **6 times more** subscribers than if they contacted a random 5% sample of the database.

Assessment Score Rankings								
Data Role=TRAIN Target Variable=TargetBuy Target Label=1								
Depth	Gain	Lift	Cumulative Lift	% Response	Cumulative % Response	Number of Observations	Mean Posterior Probability	
1	5	468.135	5.68135	66.4260	66.4260	1108	0.75790	
2	10	386.597	4.05259	47.3827	36.9043	1108	0.45343	
3	15	319.280	2.84524	4.19280	49.0220	1107	0.26503	
4	20	269.493	2.16138	3.68493	25.2708	43.0828	1108	0.17585
5	25	226.950	1.60705	3.26950	18.7895	38.2268	1107	0.12985
6	30	188.528	0.96490	2.88528	11.2816	33.7346	1108	0.10120
7	35	158.881	0.61052	2.58881	9.4765	30.2682	1108	0.08151
8	40	132.234	0.45585	2.32234	5.3297	27.1527	1107	0.06746
9	45	109.854	0.30877	2.09854	3.6101	24.5361	1108	0.05635
10	50	91.660	0.27814	1.91660	3.2520	22.4088	1107	0.04785
11	55	76.547	0.25473	1.76547	2.9783	20.6418	1108	0.04069
12	60	63.504	0.20070	1.63504	2.3466	19.1168	1108	0.03483
13	65	51.943	0.13135	1.51943	1.5357	17.7651	1107	0.02950
14	70	41.583	0.06947	1.41583	0.8123	16.5538	1108	0.02476
15	75	32.304	0.02318	1.32304	0.2710	15.4689	1107	0.02043
16	80	24.418	0.06175	1.24418	0.7220	14.5469	1108	0.01661
17	85	17.233	0.02316	1.17233	0.2708	13.7059	1108	0.01318
18	90	10.853	0.02318	1.10853	0.2710	12.9608	1107	0.01020
19	95	5.138	0.02316	1.05138	0.2708	12.2927	1108	0.00734
20	100	0.000	0.02318	1.00000	0.2710	11.6919	1107	0.00412

Balance Between Performance and Interpretability

The Logistic Regression model provides an ideal balance for banking operations:

- **High Performance:** By utilizing Validation Error for selection, the model ensures stable performance, providing a reliable ranking of customers for marketing priority.

- **Superior Interpretability:** Unlike "black-box" models (e.g., Neural Networks), the Regression model provides **Estimated Coefficients**. For instance, the positive coefficient for 'duration' allows stakeholders to clearly understand that longer consultations directly increase the probability of a subscription. This transparency is crucial for the bank's management to justify campaign strategies and train telesales staff effectively.

d. Interpretation of the Regression Model

This interpretation is based on the final model selected at **Step 11**, which identifies the key drivers behind a client's decision to subscribe to a term deposit.

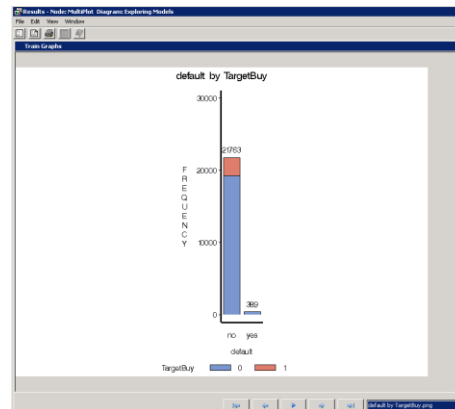
- **Factors Increasing the Likelihood of Subscription:** Based on the positive estimates in the **Analysis of Maximum Likelihood Estimates** table, the following variables are strong positive indicators:
 - **Call Duration (duration):** This is the most significant predictor (Estimate: **0.0039**). For every additional second a client stays on the phone, the likelihood of subscription increases. This suggests that engaged clients who listen to the full pitch are much more likely to commit.
 - **Account Balance (balance):** Clients with higher yearly balances have a higher probability of subscribing (Estimate: **0.000017**). Essentially, clients with more liquidity are better candidates for term deposits.
 - **Contact Method (contact):** Using **Cellular** (Estimate: **0.506**) or **Telephone** (Estimate: **0.495**) significantly increases the likelihood compared to "Unknown" contact types.
 - **Debt Status (housing=no, loan=no):** Clients who do **not** have a housing loan (Estimate: **0.357**) or a personal loan (Estimate: **0.266**) are more likely to subscribe, likely because they have fewer financial obligations and more disposable income to invest.
 - **Education (REP_education=Tertiary):** Clients with higher education levels show a higher propensity to subscribe (Estimate: **0.158**) compared to those with unknown or lower education levels.
- **Factors Reducing the Likelihood of Subscription:** The following variables have negative estimates, meaning they act as deterrents:
 - **Campaign Intensity (campaign):** The model shows a negative relationship (Estimate: **-0.078**). Each additional contact attempt made during the current campaign actually **reduces** the chance of a "Yes," indicating a point of diminishing returns where clients become annoyed or fatigued by repeated calls.
 - **Previous Outcome (REP_poutcome=Failure):** If a client was contacted previously and the outcome was a failure, they are significantly less likely to subscribe now (Estimate: **-0.808**) compared to those who successfully subscribed in the past.
 - **Job Category (REP_job=Blue-collar):** Clients in blue-collar roles have a lower likelihood (Estimate: **-0.347**) compared to the "White-collar" reference group.
 - **Marital Status (marital=married):** Married clients are slightly less likely to subscribe (Estimate: **-0.187**) compared to single individuals.
- **Reasonableness of Effects in a Bank-Marketing Setting:** These results are highly consistent with real-world marketing dynamics:
 - It is logical that **higher balances** and **lower debt** correlate with investment capacity.
 - The **negative impact of high campaign counts** highlights the "annoyance factor" in telemarketing, proving that quality of contact is more important than quantity.
 - The **strong influence of duration** confirms that the sales team's ability to hold a client's attention is a direct proxy for customer interest.
- **Informed Targeting Decisions for the Bank:** The bank can use these regression results to move from "mass calling" to a **highly targeted strategy**:
 - **Prioritize High-Potential Leads:** Focus resources on clients who have **no existing loans**, **high balances**, and a **tertiary education**, as these are the most "convertible" segments.
 - **Optimize Call Strategy:** Sales agents should be trained to maximize **engagement time**, but the marketing system should trigger an automatic "stop" after a certain number of failed **campaign contacts** to avoid brand damage and wasted effort.

- **Leverage Historical Success:** Clients with a "Success" outcome in the previous campaign are the most valuable leads and should be the first to be contacted in future rounds.

4. Neural Network Architectural:

To ensure data quality for the Neural Network, a model highly sensitive to input distributions, a visual inspection was conducted using the MultiPlot node.

- Observation: The variable default (whether a customer has credit in default) shows an overwhelming concentration in a single category, with "No" accounting for 98.24% of the observations.



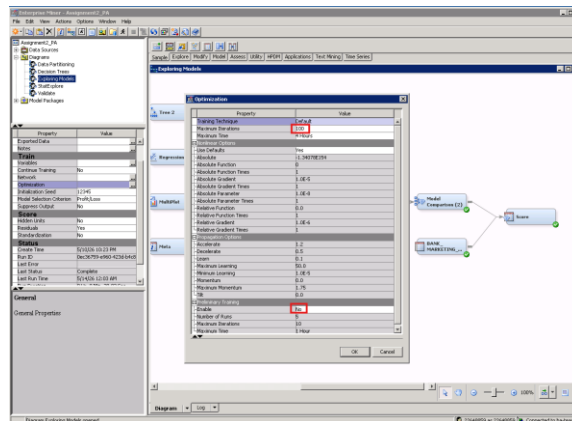
- Implication: When a nominal variable has nearly zero variance (almost all records are the same), it provides very little "information gain" for the model. In a Neural Network, such variables can lead to:
- Weight Instability: The model may struggle to assign meaningful weights to the "Yes" category due to the lack of representative samples.
- Noise: It acts more like a constant than a predictor, potentially adding unnecessary complexity without improving predictive power.

Based on the findings above, the following adjustment was made in the Metadata node (or Data Source Metadata) prior to building the Neural Network:

- Action: The Role of variables dataobs, default, and all original interval variables (e.g., balance, duration, pdays, campaign) were set to Rejected.
- Justification:
 - Statistical Sparsity & Near-Zero Variance: The variable default was excluded because 98.24% of its data falls into a single category ("No"). This lack of statistical diversity prevents the Neural Network from learning meaningful patterns, potentially leading to bias or hidden layer instability.
 - Redundancy Elimination: All original interval variables were rejected to prevent multicollinearity and mathematical redundancy. Since these variables have been processed through Impute and Replacement nodes, only their transformed counterparts (prefixed with REP_ and IMP_) are retained as Inputs.
 - Model Stability: By ensuring the Neural Network only processes "cleaned" versions of the data (where outliers are capped and missing values are treated), the model can achieve faster convergence and better generalizability on unseen data.

Name	Hidden	Hide	Role	New Role	Level	New Level	New Order	New Report
ID	N	Default	ID	ID	Nominal	Default	Default	Default
REP_order	N	Default	Rejected	Default	Interval	Default	Default	Default
M_days	N	Default	Rejected	Rejected	Binary	Default	Default	Default
REP_pdays	N	Default	Input	Default	Interval	Default	Default	Default
REP_age	N	Default	Input	Default	Interval	Default	Default	Default
REP_balance	N	Default	Input	Default	Interval	Default	Default	Default
REP_campaign	N	Default	Input	Default	Interval	Default	Default	Default
REP_day	N	Default	Input	Default	Interval	Default	Default	Default
REP_duration	N	Default	Input	Default	Interval	Default	Default	Default
REP_education	N	Default	Input	Default	Nominal	Default	Default	Default
REP_job	N	Default	Input	Default	Nominal	Default	Default	Default
REP_month	N	Default	Input	Default	Nominal	Default	Default	Default
REP_outcome	N	Default	Input	Default	Nominal	Default	Default	Default
REP_previous	N	Default	Input	Default	Interval	Default	Default	Default
Targetday	N	Default	Target	Target	Binary	Default	Default	Default
dateobs	N	Default	ID	Rejected	Interval	Default	Default	Default
age	N	Default	Rejected	Input	Interval	Default	Default	Default
balance	N	Default	Rejected	Input	Interval	Default	Default	Default
campaign	N	Default	Rejected	Input	Interval	Default	Default	Default
contact	N	Default	Input	Default	Nominal	Default	Default	Default
day	N	Default	Rejected	Input	Interval	Default	Default	Default
default	N	Default	Input	Rejected	Nominal	Default	Default	Default
duration	N	Default	Rejected	Input	Interval	Default	Default	Default
education	N	Default	Rejected	Rejected	Nominal	Default	Default	Default
housing	N	Default	Input	Default	Nominal	Default	Default	Default
job	N	Default	Rejected	Rejected	Nominal	Default	Default	Default
loan	N	Default	Input	Default	Nominal	Default	Default	Default
marital	N	Default	Input	Default	Nominal	Default	Default	Default
month	N	Default	Rejected	Rejected	Nominal	Default	Default	Default
pdays	N	Default	Rejected	Rejected	Interval	Default	Default	Default
previous	N	Default	Rejected	Default	Interval	Default	Default	Default

a. Neutral network with 3 hidden units:



A Neural Network model was developed using a Multilayer Perceptron (MLP) architecture with a single hidden layer. This model was specifically configured with **3 hidden units** (\$H11, H12, H13\$) to detect and model non-linear patterns within the bank's customer data.

Main Validation Results:

Based on the **Fit Statistics** and **Assessment Score Rankings** for the 3-unit model, the performance on the validation partition is as follows:

- Validation Misclassification Rate: 0.097 (9.7%)
- Validation Average Squared Error (ASE): 0.07
- Validation Sum of Squared Errors (SSE): 1298.73
- Cumulative Lift (at 5% depth): 5.79

Suitability Assessment:

The 3-unit Neural Network model is considered **highly suitable** for predicting customer purchase behavior in this campaign for several reasons:

- **Successful Convergence and Stability:** Unlike more complex architectures that may struggle with optimization, this model achieved stable convergence (FCONV criterion satisfied). With only **88 parameters**, it strikes a good balance between model complexity and predictive power.

- **Excellent Generalization:** The model shows no signs of overfitting. The **Average Squared Error (ASE)** is identical at **0.07** for both training and validation sets. Furthermore, the **Misclassification Rate** is consistent at **10%** across both partitions, ensuring that the model will perform reliably on new, unseen customer lists.
- **Superior Targeting Efficiency:** The **Cumulative Lift of 5.79** at the top 5% depth is very strong. This indicates that the bank can capture nearly **6 times more** actual buyers by focusing only on the top 5% of customers identified by the neural network, compared to a random calling approach.
- **Low Error Profile:** An ASE of **0.07** reflects that the predicted posterior probabilities are very close to the actual outcomes (\$0\$ or \$1\$), providing high confidence in the model's scores.

Results - Nodes Neural Network 3 Hidden Units Diagrams Exploring Models									
Fit Statistics									
Target	Target Label	Fit Statistics		Statistics Label	Train	Validation	Test		
TargetBuy		_DFT_		Total Degrees of Freedom	22152				
TargetBuy		_DFE_		Degrees of Freedom for Error	22084				
TargetBuy		_DFM_		Model Degrees of Freedom	88				
TargetBuy		_NW_		Number of Estimated Weights	88				
TargetBuy		_AIC_		Akaike's Information Criterion	9930.525				
TargetBuy		_SBC_		Schwarz's Bayesian Criterion	10635.03				
TargetBuy		_ASE_		Average Squared Error	0.068198		0.06839		
TargetBuy		_MAE_		Maximum Absolute Error	0.998938		0.998852		
TargetBuy		_DIV_		Divisor for ASE	44304		18890		
TargetBuy		_NOBS_		Sum of Frequencies	22152		9495		
TargetBuy		_RMSE_		Root Average Squared Error	0.261148		0.261515		
TargetBuy		_SSE_		Sum of Squared Errors	3021.451		1298.731		
TargetBuy		_SUMW_		Sum of Case Weights Times Freq	44304		18890		
TargetBuy		_FPE_		Final Prediction Error	0.068742		0.06839		
TargetBuy		_MSE_		Mean Squared Error	0.06847		0.06839		
TargetBuy		_RFPE_		Root Final Prediction Error	0.262187		0.261515		
TargetBuy		_RMSE_		Root Mean Squared Error	0.261688		0.261515		
TargetBuy		_AVER_		Average Error Function	0.220173		0.221381		
TargetBuy		_ERR_		Error Function	9754.525		4204.208		
TargetBuy		_MISC_		Misclassification Rate	0.098209		0.09763		
TargetBuy		_WRONG_		Number of Wrong Classifications	2133		927		

477	Data Role=VALIDATE Target Variable=TargetBuy Target Label=' '									
478										
479										
480		Depth	Gain	Lift	Cumulative Lift	% Response	Cumulative % Response	Number of Observations	Mean Posterior Probability	
481										
482										
483	5	478.832	5.78832	5.78832	67.7895	67.7895	475	0.68769		
484	10	402.433	4.26035	5.02433	49.8947	58.8421	475	0.48387		
485	15	332.027	2.91214	4.32027	34.1053	50.5965	475	0.35445		
486	20	282.194	2.32381	3.82194	27.2152	44.7604	474	0.24369		
487	25	238.094	1.61785	3.38094	18.9474	39.5956	475	0.16236		
488	30	199.708	1.07857	2.99708	12.6316	35.1000	475	0.10868		
489	35	165.870	0.62917	2.65870	7.3684	31.1372	475	0.07462		
490	40	136.960	0.34227	2.36960	4.0084	27.7514	474	0.05029		
491	45	114.216	0.32357	2.14216	3.7895	25.0878	475	0.03434		
492	50	95.303	0.25167	1.95303	2.9474	22.8728	475	0.02352		
493	55	78.359	0.08988	1.78359	1.0526	20.8884	475	0.01644		
494	60	64.568	0.12610	1.64568	1.4768	19.2733	474	0.01174		
495	65	52.733	0.10786	1.52733	1.2632	17.8872	475	0.00842		
496	70	42.076	0.03595	1.42076	0.4211	16.6391	475	0.00614		
497	75	32.840	0.03595	1.32840	0.4211	15.5574	475	0.00453		
498	80	24.888	0.05404	1.24888	0.6329	14.6261	474	0.00329		
499	85	17.538	0.00000	1.17538	0.0000	13.7653	475	0.00225		
500	90	11.105	0.01798	1.11105	0.2105	13.0119	475	0.00144		
501	95	5.254	0.00000	1.05254	0.0000	12.3268	475	0.00084		
502	100	0.000	0.00000	1.00000	0.0000	11.7114	474	0.00031		
503										

b.Change to 2 and 6 hidden units:

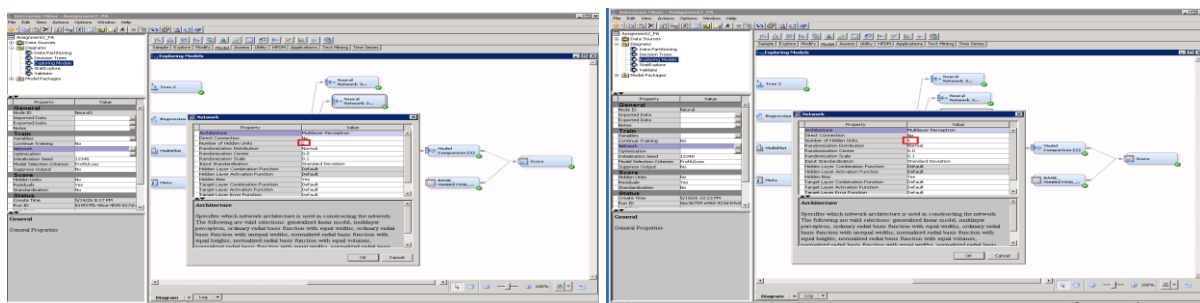
- **Performance Comparison Across Hidden Unit Settings:** To determine the optimal architecture for the neural network, the model was tested with three different hidden unit configurations: 2, 3, and 6. The performance metrics on the Validation set are summarized below:

Metric (Validation)	2 Hidden Units	3 Hidden Units	6 Hidden Units
Average Squared Error (ASE)	0.07	0.07	0.07
Misclassification Rate	10.04%	9.76%	9.77%
Number of Wrong Classifications	953	927	928
SBC (Schwarz's Bayesian)	10603.27	10635.03	11248.69
AIC (Akaike's Information)	10130.93	9930.52	9847.70

Observations:

- Justification:

- Conclusion: The 3 Hidden Unit configuration is recommended for deployment as it optimizes predictive accuracy while maintaining a streamlined structure that ensures reliable generalization for future marketing campaigns.



Target	Target Label	Fit Statistics	Statistics Label	Train	Validation	Test	Target	Target Label	Fit Statistics	Statistics Label	Train	Validation	Test
Target0	OFFT		Total Degrees of Freedom		21152		Target0	Target0	OFFT		Total Degrees of Freedom		21152
Target0	OFFT		Degrees of Freedom for Error		20001		Target0	Target0	OFFT		Degrees of Freedom for Error		20001
Target0	OFFML		Model Degrees of Freedom		59		Target0	Target0	OFFML		Model Degrees of Freedom		115
Target0	OFFML		Number of Estimated Weights		59		Target0	Target0	OFFML		Number of Estimated Weights		115
Target0	OFFML		Adjusted Information Criterion		19.130		Target0	Target0	OFFML		Adjusted Information Criterion		94.677
Target0	OFFML		Schwartz Bayesian Criterion		1983.27		Target0	Target0	OFFML		Schwartz Bayesian Criterion		1143.69
Target0	OFFML		Average Squared Error		0.00603	0.07049	Target0	Target0	OFFML		Average Squared Error		0.08872
Target0	OFFML		Maximum Absolute Error		0.08697	0.09895	Target0	Target0	OFFML		Maximum Absolute Error		0.09948
Target0	OFFML		Chi-Square Error		4.624	1.080	Target0	Target0	OFFML		Chi-Square Error		2.951
Target0	OFFML		Sum of Squares		2152	849	Target0	Target0	OFFML		Sum of Squares		2454
Target0	OFFML		Average Squared Error		0.04581	0.04487	Target0	Target0	OFFML		Root Squared Squared Error		0.20243
Target0	OFFML		Sum of Squares Error		198.01	69.1	Target0	Target0	OFFML		Sum of Squares Error		158.154
Target0	OFFML		Sum of Cross Weights Times Weq		4.624	1.080	Target0	Target0	OFFML		Sum of Cross Weights Times Weq		4.624
Target0	OFFML		Free Prediction Error		0.07049		Target0	Target0	OFFML		Free Prediction Error		0.08748
Target0	OFFML		Mean Squared Error		0.07070	0.07049	Target0	Target0	OFFML		Mean Squared Error		0.06969
Target0	OFFML		Root Free Prediction Error		0.26827		Target0	Target0	OFFML		Root Free Prediction Error		0.26803
Target0	OFFML		Root Mean Squared Error		0.26841	0.26847	Target0	Target0	OFFML		Root Mean Squared Error		0.26703
Target0	OFFML		Average Error Function		0.20001	0.22913	Target0	Target0	OFFML		Average Error Function		0.24476
Target0	OFFML		Error Fraction		100.120	420.172	Target0	Target0	OFFML		Error Fraction		94.677
Target0	OFFML		Maximum Relative Error		0.00001	0.11039	Target0	Target0	OFFML		Maximum Relative Error		0.09787
Target0	OFFML		Number of Fitting Classifications		2172	853	Target0	Target0	OFFML		Number of Fitting Classifications		2002

5. Model Comparison & ROI Translation:

To ensure a rigorous and fair comparison within the Model Comparison node, all candidate models, including the optimized Decision Tree and Logistic Regression, were integrated into the flow downstream of the final Metadata and Transformation nodes.

This architectural choice was made for two primary reasons:

- **Data Consistency:** It ensures that every model operates on the same refined feature set, where redundant identifiers (ID) and near-zero variance predictors (default) have been systematically rejected.
- **Enhanced Input Quality:** By positioning models after the Impute, Replacement, Metadata nodes, the algorithms benefit from treated outliers and handled missing values, which is particularly critical for the stability of the Regression and Neural Network coefficients.

Consequently, the final evaluation reflects the true predictive power of each algorithm under optimal data conditions rather than differences in data preparation.

a. Comparison of Modelling Approaches

- **Identification of the Best-Performing Model:** Based on the Model Comparison results, the Neural Network (3 Hidden Units) model has been identified as the Champion Model. In the Fit Statistics output, this model is explicitly marked with a "Y" in the Selected Model column, indicating it satisfies the selection criteria more effectively than the alternative Decision Tree and Regression models.
- **Justification using Validation-Based Evidence:** The selection of the Neural Network as the superior model is supported by multiple performance metrics evaluated on the Validation dataset, which ensures the model's ability to generalize to unseen data:
 - **Validation Misclassification Rate (Primary Criterion):** The Neural Network achieved the lowest misclassification rate of 0.097630 (approx. 9.76%). This is statistically superior to Tree 2 (0.098684) and the Regression model (0.099842), proving that the Neural Network provides the highest overall accuracy in predicting term deposit subscriptions.
 - **ROC Index (Discriminatory Power):** The Neural Network maintained a high ROC Index of 0.91 on both Training and Validation sets. This reflects an "excellent" ability to differentiate between potential subscribers (Events) and non-subscribers, outperforming the Decision Tree (0.86) and Regression (0.90) models.
 - **Average Squared Error (ASE):** The model recorded the lowest Validation ASE of 0.068390. A lower ASE indicates that the predicted probabilities are more closely aligned with the actual outcomes, making this model the most reliable for customer probability scoring.
 - **Event Classification (Business Impact):** In the Event Classification table for the Validation set, the Neural Network correctly identified 424 actual subscribers (True Positives), which is higher than Tree 2 (412) and Regression (397). From a business perspective, this model captures the most potential revenue-generating leads for the bank.



- Importance of Comparing Multiple Modelling Approaches: Relying on a single modelling technique is insufficient in predictive analytics for the following reasons:
 - Diverse Data Relationships: Different algorithms have unique underlying assumptions. While Regression assumes linear relationships, Neural Networks can capture complex, non-linear interactions between variables. Comparison allows us to discover which mathematical structure best fits the specific patterns within the bank's marketing data.
 - Risk of Overfitting: A model might perform exceptionally well on Training data but fail on Validation data (memorizing noise instead of patterns). Comparing multiple models helps identify the one with the best generalization capability, ensuring the bank's marketing resources are not wasted on unreliable predictions.
 - Performance-Interpretability Trade-off: There is often a trade-off between the "black-box" nature of powerful models (like Neural Networks) and the transparency of "white-box" models (like Decision Trees). By comparing them, decision-makers can determine if a significant gain in accuracy (e.g., from a Neural Network) justifies the loss of interpretability for operational implementation.

b. Scoring

- Definition and Rationale of Scoring in SAS Enterprise Miner: In SAS Enterprise Miner, Scoring is the strategic application of the mathematical rules and patterns identified during the training phase to a new, unseen dataset (Score dataset). Unlike the training process, scoring does not require a target variable; instead, it utilizes the weights and logic of the champion model, in this case, the Neural Network (Neural2) to predict outcomes for new observations. For the bank, this represents a transition from retrospective analysis to predictive execution, generating individual propensity scores for 13,564 potential customers.
- Analysis of Scoring Output and Predicted Probabilities: The scoring process generates key internal variables that quantify customer behavior. Based on the Interval Variable Summary Statistics, we can interpret the model's confidence:
 - Propensity Scoring (P_TargetBuy1 / EM_EVENTPROBABILITY): This continuous value (0 to 1) represents the likelihood of a customer subscribing.

- + Results: The mean probability across the scored set is 0.12 (12%), with a maximum propensity of 0.87.
- + Confidence Insight: The median for P_TargetBuy0 (probability of non-subscription) is 0.98, proving the model is exceptionally robust at identifying customers who are unlikely to respond. This allows for significant "noise reduction" in the campaign list.
- Classification Outcomes (I_TargetBuy / EM_CLASSIFICATION):
 - + Using a default probability cutoff of 0.5, the model successfully partitioned the 13,564 records into two actionable groups:
 - + Predicted Subscribers (Value = 1): 932 customers (6.87%).
 - + Predicted Non-Subscribers (Value = 0): 12,632 customers (93.13%).
- Strategic Campaign Planning and Business Application: The bank can utilize these scoring outputs to transform a generic "mass-marketing" effort into a Precision-Targeted Campaign:
 - Lead Prioritization & Resource Optimization: Instead of an exhaustive and costly "cold-call" approach to 13,000+ individuals, the bank should prioritize the 932 high-propensity leads identified by the model. This concentrates human resources on the segment where the conversion rate is statistically expected to be highest.
 - Tiered Outreach Strategy:
 - + High-Propensity Segment (Top 5% - 10%): Customers with scores approaching 0.87 should be routed to a premium telesales team for personalized, high-touch consultation.
 - + Medium-Propensity Segment (P25 to P75): These customers can be "warmed up" through automated, low-cost channels such as SMS or targeted email marketing before a human agent intervenes.
 - + Low-Propensity Segment: By excluding the 93.13% predicted to decline, the bank avoids operational waste and prevents "customer fatigue" (contacting people who are clearly not interested).

Estimated ROI Impact: By focusing on the 6.87% segment, the bank can achieve a significant portion of its sales targets with only a fraction of the traditional budget, drastically improving the campaign's Return on Investment (ROI).

Statistics	Label	TRAIN	VALIDATE	SCORE
MEAN	Mean	0.88	0.89	0.88
STD	Standard Deviation	0.19	0.19	0.19
N	Non Missing	22152.00	9495.00	13564.00
MIN	Minimum	0.13	0.13	0.13
P25	25th Percentile	0.86	0.87	0.86
MEDIAN	Median	0.98	0.98	0.98
P75	75th Percentile	1.00	1.00	1.00
MAX	Maximum	1.00	1.00	1.00

Variable	Numeric Value	Formatted Value	Frequency Count	Percent
I_TargetBuy	0		12632	93.1289
I_TargetBuy	1		932	6.8711

c. Recommended Model Strategy: Strategic Recommendation: Deployment of the Neural Network Model with a Hybrid Implementation Framework.

Based on a comprehensive evaluation of predictive performance and operational viability, I recommend the **Neural Network (NN)** as the primary champion model for the bank's term deposit campaign.

- **Justification for Selection (The Profit-Driven Logic):**

- **Maximizing Revenue Capture:** The Neural Network demonstrated superior **Sensitivity**, capturing **424 actual subscriptions** in the validation set, outperforming both the Decision Tree (412) and Regression (397). In a high-volume campaign, this marginal gain in "hit rate" translates into millions of euros in additional deposits, significantly increasing the campaign's ROI.
- **Superior Lead Scoring:** The model effectively identifies the top deciles of customers with the highest conversion probability. By focusing telesales resources on these high-propensity leads, the bank can achieve **Resource Optimization**, reducing unnecessary operational costs associated with low-potential calls.
- **Critical Limitation and Mitigation Strategy:**
 - **The "Black-Box" Challenge (Interpretability Tax):** A primary caution in applying the Neural Network is its inherent lack of transparency. Unlike Decision Trees, the NN does not provide explicit "If-Then" rules, which can pose challenges for regulatory compliance and non-technical stakeholder buy-in.
 - **Strategic Mitigation (The Hybrid Approach):** To balance accuracy with clarity, I recommend a **Hybrid Implementation**. The bank should use the **Neural Network's Score** to prioritize the call list (Lead Scoring), while simultaneously utilizing the **Decision Tree's Insights** (such as the critical 500-second call duration threshold and housing loan status) to train sales agents. This ensures that while the selection is driven by complex AI, the actual customer interaction remains grounded in evidence-based, human-readable sales strategies.
- **Implementation Caution:** The bank must remain vigilant regarding the **"Duration Trap."** Since call duration is only known *during* the interaction, it cannot be used for initial lead selection. The model must be deployed as a two-stage system: using demographic predictors for pre-campaign filtering and real-time behavioral triggers (Duration) for mid-call intervention.

References

Stable/modules/tree. (n.d.). Retrieved from Scikit learn: <https://scikit-learn.org/stable/modules/tree.html>